Two Engines

Memory, prediction, and the difference between them

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Week 2 — Spring 2026

"A man's memory is bound to be a distraction unless it can be discarded;
then a sleeper or a savant might think clearly."

Jorge Luis Borges, *Funes the Memorious*

Act I — Memory

Two engines, one question

We will spend the next ninety minutes inside a single question:

- what does it mean for a system to *remember*, as opposed to *predict*?
- where in your organisation is each engine appropriate?
- and what changes when the data is sparse, drifting, or adversarial?

The two engines look the same from the outside. They are not the same on the inside, and the difference will matter to every procurement decision you make this year.

The distinction is older than statistics: see Hume's Treatise on causation versus constant conjunction.

Cross-entropy, decomposed

$$\text{cross-entropy} \;=\; H(p) \;+\; \mathrm{KL}(p \,\|\, \hat{p})$$

Only the second term is yours to minimise; the first is the world's irreducible surprise.

Now switching to:

customer-urns-carousel.html

Funes remembered everything and could think nothing.

Machine learning forgets almost every-
thing — and that is why it understands.

References

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4. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006, ch. 1.
5. E. T. Jaynes, *Probability Theory: The Logic of Science*, CUP, 2003.